Graphical solutions

1 Consider the function $y = \cos(x - 30)$.

a Sketch the graph of the function for $-360 \leq x \leq 360$.

b Sketch the line $y = \frac{1}{2}$ on the same number plane.

c Hence, state all solutions to the equation $\cos(x - 30) = \frac{1}{2}$ over the domain $(-360, 360]$.

2 Consider the function $y = \sin(x - 60)$.

a Sketch the graph of the function for $-360 \leq x \leq 360.$.

b Sketch the line $y = \frac{1}{2}$ on the same number plane.

c Hence, state all solutions to the equation $\sin(x - 60) = \frac{1}{2}$ over the domain $[-360, 360)$.

3 Consider the function $y = 2\sin 4x$.

a Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.

b Sketch the line $y = 1$ on the same number plane.

c Hence, state all solutions to the equation $2\sin 4x = 1$ over the domain $[-90^\circ, 90^\circ]$. Give your answers in degrees.

4 Consider the function $y = 2\sin 2x$.

a Sketch the graph of the function for $-180^\circ \leq x \leq 180^\circ$.

b State the other function you would add to the graph in order to solve the equation $2\sin 2x = 1$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $2\sin 2x = 1$ over the domain $[-180^\circ, 180^\circ]$.

5 Consider the function $y = 3\cos 2x + 1$.

a Sketch the graph of the function for $-180 \leq x \leq 180$.

b State the other function you would add to the graph in order to solve the equation $3\cos 2x + 1 = \frac{5}{2}$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $3\cos 2x + 1 = \frac{5}{2}$ over the domain $[-180, 180]$.

6 Consider the function $y = 2\sin 3x - 3$.

a Sketch the graph of the function for $-60 \leq x \leq 60$.

b State the other function you would add to the graph in order to solve the equation $2\sin 3x - 3 = -2$.

c Sketch the graph of this fuction on the same number plane.

d Hence, state all solutions to the equation $2\sin 3x - 3 = -2$ over the domain $[-120, 120]$.

7 Consider the function $y = -2\cos 3x$.

a Sketch the graph of the function for $-120^\circ \leq x \leq 120^\circ$.

b State the other function you would add to the graph in order to solve the equation $-2\cos 3x = -1$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $-2\cos 3x = -1$ over the domain $[-120^\circ, 120^\circ]$.

Technology

8 Solve each equation for the given interval. Round your answers to one decimal place.

a $\cos 2x = 0.9$ for $0 \leq x < 360$

b $3\sin 2x = 1.2$ for $0 \leq x < 180$

c $4\cos\left(\dfrac{x}{3} + 2\right) = 1$ for $0 \leq x < 720$

9 Consider the equation $\sin\left(\dfrac{x}{2} + 60^\circ\right) = \cos\left(\dfrac{x}{2} - 60^\circ\right)$.

a State the two functions you would graph in order to solve this equation graphically.

b Sketch the graph of both of these functions using technology.

c Hence, state all solutions to the equation over the domain $[-360^\circ, 360^\circ]$.

10 Use technology to solve the following functions over the interval $[0^\circ, 360^\circ)$. Give all solutions to the nearest degree.

a $2\sin 3x - 4\sin 2x = 0$

b $2\sin x + 5\cos x = 1$

c $\cos 3x + \cos x = \sin x$

d $2\cos 3x - 3\cos 2x = 0$

e $\cos\left(\dfrac{x}{2}\right) - 4\sin 2x = 0$

f $\sin\left(\dfrac{x}{2}\right) = 7\cos 3x$

g $4\sin^5(x) = -(\cos x + 1)$

h $\cos^3(x) + \cos x = -1$

11 Use technology to solve the following equations for $0 \leq x \leq 360$. Give all solutions to two decimal places.

a $2 \sin x + 5 \cos x = 1$

b $3 \sin 2x + 2 \sin 3x = 0$

c $3 \sin(2x) = x + 1$

d $2 \cos 3x + 3 \cos 2x = 0$

e $\cos 3x + \cos x = \sin x$

f $\cos\left(\frac{x}{2}\right) - 4 \sin 2x = 0$

g $4 \sin^5 (x) = -(\cos x + 1)$

h $\cos^5 (x) + \cos x = -1$